VALENTINA AMALA RANI D

Data Analyst | Applied AI | SQL, Python, Power BI, RAG and LLM Agents
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PROFILE SUMMARY

Data analyst (MCA, 2025) who turns data into answers people can check: SQL and Power BI reporting, and AI agents that cite their sources and are tested against known answers. Built a Bank of England mortgage lending and arrears MI dashboard with a SQL analytical layer and risk classification, and a RAG agent over HDFC Bank annual reports that scored 17/17 on a 17-question test set (plain RAG: 14/17). Focused on data quality, consistent metric definitions and clear findings for non-technical readers.

TECHNICAL SKILLS

Languages and query: SQL (window functions, CTEs, LAG, RANK), Python (Pandas, NumPy, scikit-learn)

BI and reporting: Power BI (DAX), Tableau, Streamlit, Matplotlib, Seaborn, Excel, Power Automate (basic flows)

Data and databases: Snowflake, MySQL, SQLite, MongoDB, ETL, data cleaning and validation, data profiling

AI and machine learning: RAG, Gemini API (function calling), ChromaDB, hybrid search (embeddings and BM25), FastAPI, Docker, Linear Regression, Random Forest, VADER sentiment analysis, model evaluation

Other: Flask, Git and GitHub, process documentation, stakeholder communication

PROJECTS

Financial Filings Analyst Agent (RAG and AI agent) | *Python, Gemini API, ChromaDB, FastAPI, Docker* **Nov 2026**

Live demo | Code

- Built a retrieval-augmented agent over HDFC Bank's FY25 and FY26 annual reports: PDF chunking, local ChromaDB embeddings, hybrid semantic and BM25 search, Gemini function calling. Every answer cites its report and page; out-of-scope questions are refused.
- Added calculator and year-comparison tools so percentage changes and unit conversions are computed in code, not by the language model.
- Wrote a 17-question evaluation set with automated scoring: agent 17/17 vs 14/17 for plain RAG (small set, single runs). Deployed as a FastAPI service with a web interface and per-IP rate limiting.

Bank of England Mortgage Lending MI Dashboard | *Python, Pandas, SQL, Power BI* **2026**

Code

- Built a Python and Pandas ETL process to clean and validate quarterly lending and arrears data (2020 to 2025) into a SQLite analytical layer feeding Power BI.
- Wrote five SQL queries (window functions, CTEs, LAG, RANK), defined NORMAL / ELEVATED / HIGH arrears risk rules, and delivered a 3-page Power BI dashboard.
- Finding: lending fell 33% in 2020-Q2 and peaked at £89.1bn in 2021-Q2; arrears crossed into HIGH risk in 2023-Q4 and peaked at 1.31% in 2024-Q2.

Indian Job Market Intelligence | *Python, Pandas, scikit-learn, Streamlit* **Feb 2026**

Live app | Code

- Cleaned and analysed 97,000+ raw job postings (29,000+ after cleaning) with IQR outlier removal and feature engineering; compared Linear Regression and Random Forest using MAE, RMSE and R2 (best R2 0.17: job title and state alone explain little of salary variation).
- Delivered a live Streamlit dashboard with filters and a salary prediction tool; median advertised salary rises from about INR 2.6 lakh (0 to 2 years) to INR 10 lakh (10+ years).

UK Tourism Forecasting | *Python, Prophet, Pandas* **Nov 2025**

Code

- Forecast UK international tourist arrivals with Prophet on World Bank data (1995 to 2018): about 42.6 million by end of 2027 (range 40.2 to 45.0 million), with the limitation that the data excludes the COVID-19 shock.

MCA Final Year Project: Recommendation System | *Python, VADER, collaborative filtering* **Dec 2024 - Apr 2025**

- Built a recommendation system combining VADER sentiment analysis with collaborative filtering; covered requirements, database design, development, testing and documentation.

EDUCATION

Master of Computer Applications (MCA), Jamal Mohamed College, Trichy | CGPA 8.29 / 10 **May 2025**

Bachelor of Computer Applications (BCA), Holy Cross College, Trichy | CGPA 7.78 / 10 **May 2023**

CERTIFICATIONS

Google Data Analytics Professional Certificate | Data Analytics Job Simulation, Deloitte Australia (Forage) | Python Programming, NPTEL | MEAN Stack Web Development